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COURSE NAME: ANALOG AND DIGITAL COMMUNICATION (ECA0802)

EX NO: 3b

POWER SPECTRAL DENSITY (PSD) ANALYSIS OF PHASE
MODULATED (PM) SIGNALS USING MATLAB

Objectives

- i) To generate a Phase Modulated (PM) signal in MATLAB using specified carrier and message signal parameters.
- ii) To compute the frequency spectrum of the PM signal using the Fast Fourier Transform (FFT).
- iii) To calculate and plot the Power Spectral Density (PSD) of the PM signal using MATLAB.
- iv) To identify the carrier component and sideband frequencies present in the PM spectrum.
- v) To analyze the effect of phase modulation on the spectral distribution and bandwidth of the transmitted signal.
- vi) To investigate the influence of modulation index on the Power Spectral Density characteristics of the PM signal.

PROGRAM (MATLAB CODE)

```
clc;  
clear;  
close all;  
% Parameters  
Am = 1;  
fm = 100;  
Ac = 5;  
fc = 1000;  
fs = 20*fc;  
kp = pi/2;
```

```

% Message Amplitude (V)
% Message Frequency (Hz)
% Carrier Amplitude (V)
% Carrier Frequency (Hz)
% Sampling Frequency (Hz)
% Phase Sensitivity (rad/V)
% Time Vector
t = 0:1/fs:4/fm;

% Message Signal
m = Am*sin(2*pi*fm*t);

% PM Signal
pm = Ac*cos(2*pi*fc*t + kp*m);

% Phase Deviation
beta = kp*Am;

% FFT
N = length(pm);
f = linspace(-fs/2,fs/2,N);
PM_FFT = abs(fftshift(fft(pm)))/N;

% Power Spectral Density
PSD = (abs(fftshift(fft(pm))).^2)/(N*fs);

% Approximate Bandwidth
BW = 2*(beta+1)*fm;

fprintf('Phase Deviation = %.2f radians\n',beta);
fprintf('Estimated Bandwidth = %.2f Hz\n',BW);

% Plotting
figure('Name','PSD Analysis of PM Signal','NumberTitle','off');
subplot(2,2,1);
plot(t,m);
title('Message Signal');

```

```

xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(2,2,2);
plot(t,pm);
title('Phase Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(2,2,3);
plot(f,PM_FFT);
title('Frequency Spectrum of PM Signal');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
grid on;
subplot(2,2,4);
plot(f,PSD);
title('Power Spectral Density');
xlabel('Frequency (Hz)');
ylabel('PSD');
grid on;

```

OUTPUT:

